



Measurement & Motion

Measurement

A measurement is the comparison of the unknown quantity with some standard quantity of the same kind.

- (1) A number
- (2) A unit

Units

A unit is the smallest quantity in terms of which other quantities can be measured.

Need for standard units of measurement

- (1) It is also important that the unit must be accurate and it must not change with respect to a person or time.
- (2) It is also important to have units that are understood by all.

Measurement of length

Length is a fundamental quantity and it is commonly used to measure the distance between two points.

Systems of measuring length

Metric system

millimetres(mm)
centimetres (cm)
meters(m),
kilometres (km)

English system

inches(in.),
feet (ft),
yards (yd),
miles(mi)

Conversion factors for length

$$1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm}$$

$$1 \text{ km} = 1000 \text{ m} = 100000 \text{ cm} = 1000000 \text{ mm}$$

$$1 \text{ cm} = 10 \text{ mm}$$

$$1 \text{ A}^\circ = 10^{-10} \text{ m}$$



Devices use measure length

Metre stick

Measuring Tape

Correct procedure to measure length

For measuring the length of an object, you must choose a suitable device.



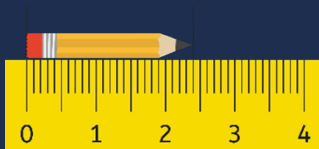
You cannot measure the circumference or girth of a tree, the size of your waist using a metre scale.

Measuring tape is more suitable for this.

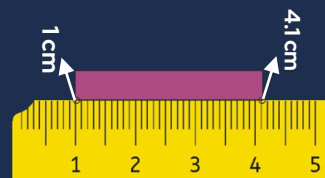


Care to be taken while measuring a length

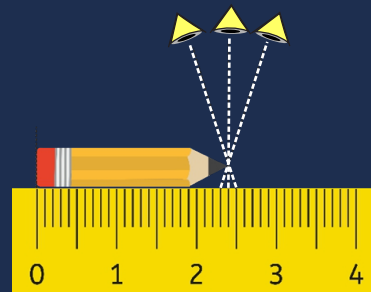
(1) Place the scale in contact with the object along its length



(2) In some scales, the ends may be broken. You may not be able to see the zero mark clearly.



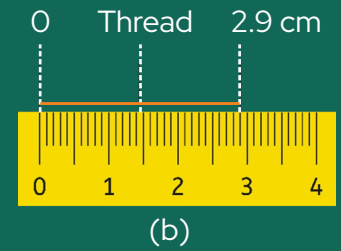
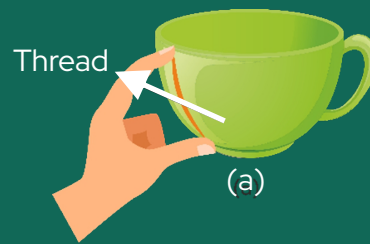
(3) Correct position of the eye is also important for taking measurement.





Measuring length of a curved line or surface

Use a string or thread and a ruler to measure the length of a curved line or surface indirectly



Use of our body parts for measurement of length

In ancient days, long before measuring instruments were invented, people actually used different parts of their body to measure length.

(1) Span (or hand span)



(2) Thumb



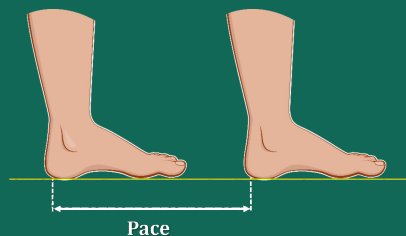
(3) Fathom



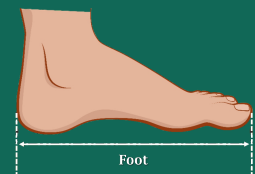
(4) Cubit



(5) Pace



(6) Foot



Limitations of our senses and body parts

- (1) accuracy of measurement
- (2) reliability of measurement
- (3) uniformity of measurement



MOTION

The act or process of changing position relative to some fixed point during a period of time.



Rest

When an object doesn't change its position with respect to surrounding in a passage of time then it is said to be in rest.



Types of Motion

Translational

Oscillatory

Rotational

Circular

Periodic

Rectilinear

Curvilinear

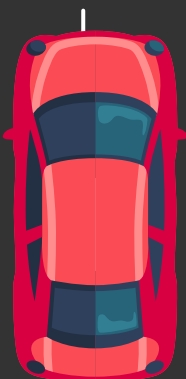
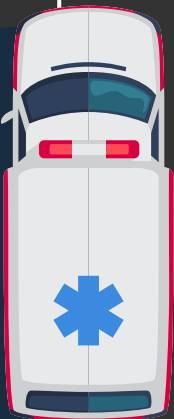
Translational motion (or translatory motion)

Motion of a body in which all the points in the body follow parallel paths

Example :- Bullet Which Gets Fired, A Car Moving In A St. Line

Rectilinear





Curvilinear



Circular Motion

When an object moves along a circular path, this type of curvilinear motion is called circular motion.

Example

- Merry-Go-round
- The Moon Orbiting Around Earth.



Rotational Motion (Rotatory Motion)

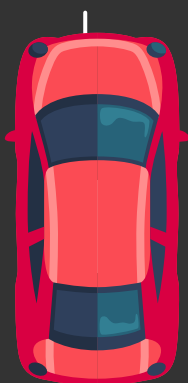
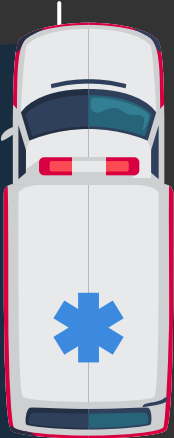
Motion of a body turning about an axis is called rotational motion.

Example

- Earth Rotating About Its Axis
- A Wheel Rolling Down A Hill.



Rolling motion is a combination of translational and rotational motion.



Periodic Motion

A motion that occurs when an object moves in a repeated pattern (a cycle) over equal periods of time is called a periodic motion.

Example

- Rocking Chair
- A Bouncing Ball

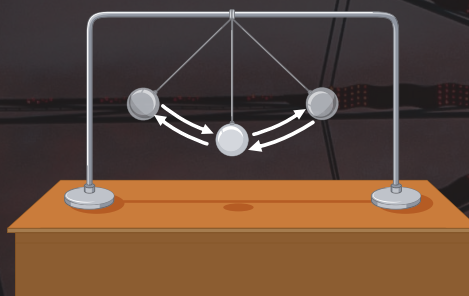


Oscillatory Motion

A motion that occurs when an object moves to and fro about its mean position over equal periods of time is called an oscillatory motion or vibratory motion.

Example

- Vibrating String
- Flapping Of Wings



Story of transportation

